Housing Allocation, Transfers, and Fertility

An illustrative model

Discussion draft for Tommaso De Santo September 8, 2026

Borrowing limits can keep housing with old owners even when young households would gain more from the space. We compare the market allocation with an equally weighted utilitarian benchmark. We then ask whether taxes and transfers can improve that allocation while households continue to choose through markets, and how the resulting changes in housing and consumption affect fertility.

1 Environment

Households. At date t there are Y_t young and O_t old households. A young household has liquid wealth $b_i > 0$, income $y_i^y > 0$ when young, and income $y_i^o \geq 0$ when old. The triple (y_i^y, b_i, y_i^o) has distribution F in each entering cohort. Future income is known. Households choose completed fertility once and retain their tenure when old.

Preferences. Each child uses $\chi > 0$ units of goods and $\kappa > 0$ units of space. Total nondurable expenditure is c and housing is h; the bundles left for adults are:

$$x = c - \chi n > 0, \qquad s = h - \kappa n > 0. \tag{1}$$

Young and old utility are:

$$u_t^y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n, \tag{2}$$

$$u^o(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e. \tag{3}$$

All preference weights are positive. Households discount future utility by $\beta > 0$. Fertility $n > 0$ is continuous; the superscript 2 labels old-age quantities. The estate e consists of financial assets and the proceeds from selling housing at death. It enters the parent's utility but does not determine an entering household's wealth b_i.

A household draws an ownership preference ξ_i before making its choices. The draw is independent of its endowments and logistic with location $\bar{\xi}$ and scale $\sigma_\xi > 0$.

Housing and financial markets. The housing stock is $\bar{H} > 0$. Rental units satisfy $h \leq h_R^{\max}$ and owner units satisfy $h \leq h_O^{\max}$, where $0 < h_R^{\max} < h_O^{\max}$. Goods and bonds trade with the rest of the world. The bond price is $q \in (0, 1)$ and its gross return is $R_f = 1/q$. House prices are $P_t > 0$. Owners pay property tax at rate τ^p. Competition among rental intermediaries gives:

$$u_t \equiv q r_t = (1 + q\tau^p) P_t - q P_{t+1} > 0. \tag{4}$$

Here r_t is rent paid at the end of the period, and u_t is its value at the beginning. Property-tax revenue is rebated equally to young and old households; T_t denotes the rebate at the beginning of the period.

Home finance. Current income and liquid wealth are available at purchase. A mortgage finances at most the share $\phi_t \in (0, 1)$ of the house price. Principal and accumulated interest are repaid on entering old age. Unsecured borrowing is unavailable. Old owners can buy or sell housing within the owner size limit, using their income and wealth without new borrowing. Thus an old owner's choice is not bounded by the size of its previous home.

Demography. A child produces $\nu > 0$ young households next period, after survival and household formation. Cohort masses satisfy:

$$Y_{t+1} = \nu \bar{n}_t Y_t, \qquad O_{t+1} = Y_t, \tag{5}$$

where $\bar{n}_t$ is average fertility among the young. Population here counts adult households.

2 Household choices

Young renters. Financial wealth on entering old age is a'. Conditional on renting, the young household solves:

$$W_t^R(i) = \max_{c,h,n,a'} u_t^y(c,h,n) + \beta V_{t+1}^R(a';i),$$
$$c + qa' + u_t h = y_i^y + b_i + T_t, \qquad a' \geq 0, \quad h \leq h_R^{\max}. \tag{6}$$

The renter pays for current housing services and saves for old age.

Young owners. For an owner, a' is financial wealth net of mortgage repayment. Its problem is:

$$W_t^O(i) = \max_{c,h,n,a'} u_t^y(c,h,n) + \beta V_{t+1}^O(a',h;i),$$
$$c + qa' + (1+q\tau^p)P_t h = y_i^y + b_i + T_t,$$
$$qa' + \phi_t P_t h \geq 0, \qquad h \leq h_O^{\max}. \tag{7}$$

To see the mortgage directly, let d be principal borrowed at purchase and k the amount invested in bonds. Then $0 \leq d \leq \phi_t P_t h$, $k \geq 0$, $a' = (k-d)/q$, and the budget is $c + k + (1+q\tau^p)P_t h = y_i^y + b_i + T_t + d$. Eliminating k, d gives (7).

Old households. An old owner has net financial wealth a and title to H units of housing. Let $a^e \geq 0$ be financial saving during old age. The estate is:

$$e = \begin{cases} q^{-1}a^e, & \text{renter,} \\ q^{-1}a^e + P_{t+1}h^2, & \text{owner.} \end{cases} \tag{8}$$

The retained house is sold at death, at the end of old age. The two old-age problems are:

$$V_t^R(a;i) = \max_{c^2,h^2,e} u^o(c^2,h^2,e),$$
$$c^2 + qe + u_t h^2 = a + y_i^o + T_t, \quad h^2 \leq h_R^{\max}, \tag{9}$$
$$V_t^O(a,H;i) = \max_{c^2,h^2,e} u^o(c^2,h^2,e),$$
$$c^2 + qe + u_t h^2 = a + P_t H + y_i^o + T_t,$$
$$h^2 \leq h_O^{\max}, \qquad e \geq P_{t+1}h^2. \tag{10}$$

The owner's estate restriction is equivalent to $a^e \geq 0$. Its cash budget before substituting for the estate is $c^2 + a^e + (1+q\tau^p)P_t h^2 = a + P_t H + y_i^o + T_t$. Income, liquid wealth, and sale proceeds therefore enter the same budget.

Tenure. The household owns if $W_t^O(i) + \xi_i \geq W_t^R(i)$. Its ownership probability is:

$$\pi_t^O(i) = \frac{\exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}{\exp\{W_t^R(i)/\sigma_\xi\} + \exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}. \tag{11}$$

The taste draw affects tenure but not choices conditional on tenure.

3 Equilibrium and the welfare comparison

Definition 1. *An equilibrium consists of household choices, tenure probabilities, prices, rebates, and cohort masses satisfying the household problems, rental pricing, and demographic equations above. Housing and the property-tax budget clear:*

$$Y_t \mathbb{E}_t h^y + O_t \mathbb{E}_t h^o = \bar{H}, \qquad (Y_t + O_t)T_t = q\tau^p P_t \bar{H}. \tag{12}$$

The old distribution is generated by the preceding cohort's choices.

At a positive stationary equilibrium, $Y = O = N$, $\bar{n} = 1/\nu$, and $N = \bar{H}/\mathbb{E}[h^y + h^o]$.

For the stationary results assume $0 \le \tau^p < 2$. At stationarity, let $w = y^y + b + T$ be current cash and $v = y^o + T$ be old income including the rebate. Write:

$$\begin{gathered} p = (1 - q + q\tau^p)P, \qquad L = (1 - \phi + q\tau^p)P, \\ z = a' + Ph + v, \qquad K = 1 + \gamma + \omega_B, \qquad D = 1 + \alpha + \vartheta + \beta K. \end{gathered} \tag{13}$$

Here p is the cost of housing services, L is the cash required per unit of owner housing, and z is the owner's resources on entering old age. The young owner's two financial restrictions become:

$$c + ph + qz = w + qv, \qquad c + Lh \le w. \tag{14}$$

The second inequality is the original mortgage limit. If old housing and the estate restriction are slack, $V(z) = K \log z + C$, with $c^2 = z/K$, $h^2 = \gamma z/(Kp)$, and $e = \omega_B z/(Kq)$.

For the direct housing comparison, hold fertility and tenure fixed at their market choices. Give each living household weight one on its remaining utility:

$$\mathcal{W}_t = \int \left[u_t^y + \beta V_{t+1} + \xi_i \mathbf{1}\{\text{owner}\}\right] \mathrm{d}M_t^Y + \int u_t^o \, \mathrm{d}M_t^O. \tag{15}$$

The measures count households. Thus equal weights do not mean equal weights on the two age-group averages. This is the proposed welfare convention for this note. Equal weights on utility as measured at birth would instead put β on the initial old; Appendix F explains the difference.

The direct planner can assign housing and waive individual borrowing limits, while respecting the housing stock, tenure-specific size limits, estates, and aggregate resources. A transfer authority instead announces lump-sum payments and lets households choose through markets under the original restrictions. We first compare allocations that leave all later cohorts unchanged, so their welfare cancels. Neither comparison attaches a social value to additional births.

4 Housing allocation

The relevant young owners have a strictly binding mortgage limit. They are below their housing cap, and their old-age housing cap and estate floor are slack. These properties follow from explicit income and capacity inequalities in Appendix A; they need not be assumed from a computed equilibrium.

Proposition 1 (Direct utilitarian improvement)**.** *Consider a positive stationary equilibrium with a positive share of these owners. If $\beta \ge q$, a small reallocation of housing from their current old counterparts to the young raises $\mathcal{W}_t$. Consumption, fertility, tenure, estates, all future real allocations, and net external assets can remain unchanged.*

Proof. Let $m = 1/c^2$ be the marginal utility of cash when old, and let $\mu > 0$ be the multiplier on the young owner's cash restriction in (14). The young and old first-order conditions imply:

$$\frac{\alpha}{s} = \frac{\beta}{q} pm + \mu L, \qquad \frac{\gamma}{h^2} = pm. \tag{16}$$

Stationarity supplies an old owner with the same endowments and old resources. The difference in housing marginal utilities is therefore:

$$\boxed{\frac{\alpha}{s} - \frac{\gamma}{h^2} = \left(\frac{\beta}{q} - 1\right) pm + \mu L > 0.} \tag{17}$$

Giving the young ε units and taking the same amount from the old changes welfare by $\alpha \log(1 + \varepsilon/s) + \gamma \log(1 - \varepsilon/h^2)$, which is positive for small $\varepsilon > 0$. Appendix B verifies the financial settlement. Stationary young and old owner measures coincide, so the comparison integrates over heterogeneous endowments. □

Young households value the same space more because current resources are scarce relative to their resources when old. At $\beta = q$, the entire gap in (17) is due to the mortgage constraint. At $\beta > q$, the household's relative valuation of the two ages also contributes under the chosen social weights. The old lose housing utility; compensation is not required by this utilitarian criterion.

This proves that the market allocation does not maximize the planner's objective. It also identifies an improving direction toward young housing. It need not identify the direction of the global optimum when other households and binding physical caps are included. For the conditional housing optimum, hold consumption, children, and estates fixed, and write $C_n = \kappa \int n_i \, \mathrm{d}M^Y$ for space used by children. If the unconstrained allocation respects all retained caps and estate floors, young aggregate housing is:

$$H_Y^* = C_n + \frac{\alpha Y}{\alpha Y + \gamma O}(\bar{H} - C_n). \tag{18}$$

At stationarity, this exceeds equilibrium young housing if all old housing caps and estate floors are slack and the hypotheses of Proposition 1 hold. Appendix B gives the aggregate comparison and the solution with active caps.

Along a transition, stationarity can no longer identify current old marginal utility with the young household's future old marginal utility. The exact local test becomes:

$$p_t \left(\frac{\beta}{q} m_{i,t+1} - m_{j,t}\right) + \mu_{i,t} L_t > 0, \tag{19}$$

where $p_t = u_t$ and $L_t = (1 - \phi_t + q\tau^p)P_t$. This applies at a date on any existing path; it requires no convergence assumption. The stationary parameter condition alone does not sign it along every path.

5 Taxes and transfers through markets

A grant can relax the young household's cash shortage. A tax on an old owner can release housing. Their demands must still clear the market, and the program must finance both current and promised payments.

There is a simple constrained utilitarian comparison that preserves the original timing of all young choices.

Proposition 2 (Redistribution among old owners)**.** *Suppose a positive stationary equilibrium has two positive groups of old owners with slack housing caps and estate floors. If every household in one group has more old-age resources than every household in the other, a small, unanticipated one-time balanced cash transfer from the richer group to the poorer raises utilitarian welfare. Every household chooses optimally under the original constraints, and the original aggregate housing, fertility, and price path are preserved.*

Proof. At the same prices, old demands are proportional to resources:

$$(c^2, h^2, e) = z(1/K, \gamma/(Kp), \omega_B/(Kq)). \tag{20}$$

A balanced transfer therefore preserves all three aggregates. Select equal positive submeasures from the two groups with uniform strict margins. For a matched unit of mass with $z_L < z_H$, transferring b raises welfare at the rate:

$$K\left[\frac{1}{z_L + b} - \frac{1}{z_H - b}\right] > 0 \qquad \text{for } 0 < b < (z_H - z_L)/2. \tag{21}$$

Choose a smaller transfer if needed to preserve the slack caps. It is a one-time intervention for current old households, so current young face the same prices and lifetime resources and retain all their choices. Estates do not determine entrant wealth, so later cohorts are also unchanged. □

The gain comes from redistributing wealth within old age. Young housing is unchanged. The condition on resource dispersion also has a simple primitive case: when $\phi = q$, owners with binding mortgages enter old age with $z = y^o + T$. Two separated old-income groups satisfying Appendix A therefore suffice. This result needs no restriction on β/q.

Moving housing toward young households through transfers requires more. The following construction is conditional on fertility and tenure having already been chosen. One interpretation is an unexpected intervention after those commitments and before housing purchase. This additional timing restriction is explicit: it is not yet a theorem for the original simultaneous fertility, tenure, and housing choices.

Proposition 3 (A transfer improvement with committed fertility)**.** *Start from the stationary equilibrium in Proposition 1. Suppose:*

$$\boxed{\phi \geq q, \qquad \alpha(1 + \omega_B) \leq \gamma \frac{1 - \phi + q\tau^p}{1 - q + q\tau^p}.} \tag{22}$$

Suppose also that a positive group of other old owners is strictly at its housing cap. A small program of current and next-period grants to selected young owners, financed by taxes on current old owners, raises utilitarian welfare and moves housing toward the young. Households optimize conditional on their committed fertility and tenure; all private financial restrictions remain. The government saves current revenue to pay its next-period grants.

The preference restriction makes the grants' remaining funding requirement nonnegative after taxing the old owners who release housing. The additional wealthy old owners cover that requirement without changing their housing, because their cap remains binding. Their marginal utility of cash is automatically lower than that of the uncapped old donors:

$$m_F < \frac{\gamma}{p h_O^{\max}} < m_i. \tag{23}$$

Appendix C gives an income bound producing this group and proves that the restrictions can hold jointly. This construction includes redistribution from households with low marginal utility of consumption; its gain cannot all be attributed to housing.

Here is the policy for one selected young owner. Put $\lambda = \beta K/(qz)$. For a small increase ε in adult space, choose:

$$s_\varepsilon = s + \varepsilon, \qquad x_\varepsilon = \frac{L}{\alpha/(s+\varepsilon) - (p-L)\lambda}. \tag{24}$$

The current grant G and next-period grant J are:

$$G = x_\varepsilon - x + L\varepsilon, \qquad qJ = (p-L)\varepsilon. \tag{25}$$

Both are nonnegative. These are fixed payments, calculated from the baseline, not subsidies that vary with the household's subsequent choice. The household voluntarily chooses (24), remains at its mortgage limit, and enters old age with its original resources z.

An uncapped old owner's housing demand is gz, where $g = \gamma/(Kp)$. Tax its matched old counterpart ε/g, so that housing falls by exactly ε. The wealthy capped group pays the residual:

$$R = G + qJ - \varepsilon/g = x_\varepsilon - x + p\varepsilon - \varepsilon/g \geq 0. \tag{26}$$

Thus housing clears at its original price. Current net revenue is qJ; saving it pays J next period. The original old-age resources and choices of the young are preserved, and later cohorts face the original economy. The initial old leave smaller estates, which are included in their welfare.

This establishes one feasible transfer improvement. It does not characterize a uniform age-transfer rule or the optimal transfer schedule. Allowing fertility to respond changes housing demand and future cohort masses. Appendix E gives a different construction that lets individual fertility and tenure remain private choices while preserving total births. The next section distinguishes these welfare comparisons from a policy that raises aggregate fertility.

6 Fertility on the adjustment path

Children use both goods and space. The fertility condition is:

$$\frac{\vartheta}{n} = \frac{\chi}{c - \chi n} + \frac{\alpha\kappa}{h - \kappa n}. \tag{27}$$

Additional housing lowers the space cost of another child; a fall in consumption raises its goods cost. This distinction matters even when the household gains utility.

Proposition 4 (A gift and a dated fertility comparison)**.** *At fixed tenure, prices, and old-age transfers, a current cash gift strictly raises fertility. Housing rises until its physical cap binds. The result allows either mortgage regime and all old-age housing and estate constraints. The original pair of current and old-age grants in (25) also strictly raises a treated owner's fertility at the fixed price-and-rebate path, after all its choices are reoptimized.*

More generally, compare two privately chosen bundles at the same fertility preference ϑ. If the new bundle can accommodate baseline fertility n_0, then:

$$\boxed{n_1 \geq n_0 \quad \Longleftrightarrow \quad \frac{\chi}{c_1 - \chi n_0} + \frac{\alpha\kappa}{h_1 - \kappa n_0} \leq \frac{\chi}{c_0 - \chi n_0} + \frac{\alpha\kappa}{h_0 - \kappa n_0}.} \tag{28}$$

This test applies at a date on two existing equilibrium paths, without requiring either path to converge.

A rise in both consumption and housing is sufficient for higher fertility. If one falls, (28) states exactly how much the other must compensate. Appendix D proves the gift result across constraint regimes. A gift paid for by taxing today's old is different from a loan repaid by the young recipient: anticipated repayment can reverse its fertility response.

Tenure changes. A common gift raises conditional fertility in both tenures, but households may switch between them. For each endowment type, mean fertility obeys the exact identity:

$$\Delta \bar{n}_i = \pi_1^O \Delta n_i^O + (1-\pi_1^O)\Delta n_i^R + (\pi_1^O - \pi_0^O)(n_{i0}^O - n_{i0}^R). \tag{29}$$

The last term can be negative. A grant available only to buyers raises ownership at fixed prices; if, for every affected endowment type, owners initially choose at least as many children as renters, both contributions are nonnegative. This is an additional tenure comparison, not a consequence of utilitarian improvement.

The same grant pair raises fertility. Give the household exactly the grants in (25). Let $W_\varepsilon(n)$ be its value after optimizing all other choices conditional on fertility n. At baseline fertility n_0, the conditional optimum has $x_\varepsilon > x_0$ and $s_\varepsilon > s_0$. The envelope theorem gives:

$$W'_\varepsilon(n_0) = \chi\left(\frac{1}{x_0} - \frac{1}{x_\varepsilon}\right) + \alpha\kappa\left(\frac{1}{s_0} - \frac{1}{s_\varepsilon}\right) > 0. \tag{30}$$

The profiled value is concave, so the new optimal fertility is strictly higher. This includes the promised old-age grant and all household reoptimization. It does not require consumption and housing both to rise at the final, freely chosen fertility.

This positive treatment effect does not preserve the market implementation. For example, when $\phi = q$ and $\gamma = \alpha(1+\omega_B)$, the balanced transfer in Appendix C clears housing at fixed fertility. With fertility freely chosen, the difference between young and old housing responses to cash instead is:

$$\frac{\vartheta(\kappa p - \alpha\chi)}{p(\chi + \kappa p)(1+\alpha)(1+\alpha+\vartheta)}. \tag{31}$$

It can have either sign. Even if it is zero, higher fertility changes the next cohort's housing demand. A policy that raises aggregate fertility therefore needs a new demographic equilibrium path. The separate construction in Appendix E avoids fixing individual fertility by offsetting births within the young cohort; total fertility remains unchanged.

The intended transition. Start from a stationary economy. An exogenous decline in ϑ_t initiates demographic adjustment. At a later date, compare a housing policy with continued adjustment without it, from the same inherited state and with the same preference shocks. The initial fertility decline is part of the experiment, rather than something the housing mechanism must explain.

For two existing paths sharing Y_{t_0}, cohort accounting gives:

$$\frac{Y_t^{\text{policy}}}{Y_t^{\text{baseline}}} = \prod_{s=t_0}^{t-1} \frac{\bar{n}_s^{\text{policy}}}{\bar{n}_s^{\text{baseline}}}, \qquad O_{t+1}^j = Y_t^j. \tag{32}$$

This orders populations at finite dates if fertility is ordered along the path. Convergence is unnecessary for either this identity or the dated fertility test. Determining equilibrium consumption, housing, and prices still requires an equilibrium path, including the policy's fiscal response.

If both paths converge to positive stationary equilibria, both fertility limits equal $1/\nu$. Their limiting adult-household populations satisfy:

$$\frac{(Y+O)^{\text{policy}}}{(Y+O)^{\text{baseline}}} = \frac{\mathbb{E}[h^y + h^o]^{\text{baseline}}}{\mathbb{E}[h^y + h^o]^{\text{policy}}}. \tag{33}$$

A Primitive conditions for the stationary result

The following sufficient conditions produce the equilibrium and constrained owner group in Proposition 1. They are conservative analytical bounds, with no equilibrium price or multiplier as an input. Assume bounded endowments, $w_{0i} = y_i^y + b_i \geq \underline{w} > 0$, $v_{0i} = y_i^o \geq 0$, and $0 \leq \tau^p < 2$. Define:

$$\begin{aligned} d_p &= 1 - q + q\tau^p, \quad d_L = 1 - \phi + q\tau^p, \quad d_{\max} = \max\{d_p, d_L\}, \\ \bar{M}_0 &= \mathbb{E}(w_0 + qv_0), \quad \bar{T} = \frac{\tau^p \bar{M}_0}{(1-q)(2-\tau^p)}, \quad \bar{M} = \bar{M}_0 + (1+q)\bar{T}. \end{aligned} \tag{34}$$

Require enough potential fertility when housing is inexpensive:

$$h_R^{\max} > \kappa/\nu, \qquad \vartheta\nu > \chi D/\underline{w} + \frac{\alpha\kappa}{h_R^{\max} - \kappa/\nu}. \tag{35}$$

Then a positive stationary equilibrium exists, and every such equilibrium has $0 \leq T \leq \bar{T}$ and $P_- < P < P_+$, where:

$$P_- \equiv \frac{\alpha\underline{w}}{Dd_{\max}h_R^{\max}}, \qquad P_+ \equiv \frac{\nu\bar{M}}{d_p\kappa}. \tag{36}$$

To verify existence, combine the original budgets in either tenure:

$$x + ps + (\chi + p\kappa)n + qc^2 + qph^2 + q^2e = w + qv. \tag{37}$$

Scaling the first-order conditions gives $\lambda(w+qv)+\mu w \leq D$, while $1/x = \lambda+\mu$; hence $x \geq w/D \geq \underline{w}/D$. Below P_-, the housing condition $\alpha x/s \leq d_{\max}P$ when uncapped, or the binding cap itself, implies $h \geq h_R^{\max}$. Equation (27) and (35) then give fertility above replacement for all conditional choices. Above P_+, (37) gives mean fertility below replacement for every $T \leq \bar{T}$.

The rebate map, writing $\bar{h} = \mathbb{E}(h^y + h^o)$, satisfies:

$$\mathcal{T}(P, T) = q\tau^p P\bar{h}/2 \leq \frac{\tau^p}{2d_p}\{\bar{M}_0 + (1+q)T\} \leq \bar{T}. \tag{38}$$

The final inequality uses $2d_p - \tau^p(1+q) = (1-q)(2-\tau^p) > 0$. Conditional allocations and tenure probabilities are continuous. On the compact rectangle of prices and rebates, move the price in the direction of $\nu\bar{n} - 1$, clipping it to $[P_-, P_+]$, and update the rebate by $\mathcal{T}$. A Brouwer fixed point has an interior price by the strict boundary signs, and therefore replacement fertility. Setting $N = \bar{H}/\bar{h}$ completes the stationary equilibrium. This argument establishes existence, not uniqueness or transition stability.

For the constrained owner group, put:

$$\ell = d_L/d_p, \qquad C_{\min} = 1 + \alpha\ell + \vartheta\min\{1, \ell\}, \qquad k = D/C_{\min} - 1. \tag{39}$$

On a positive-F-mass set $\mathcal{S}$, require:

$$qv_{0i} > kw_{0i} + \max\{k - q, 0\}\bar{T}. \tag{40}$$

For $M_{\mathcal{S}} \geq \sup_{i\in\mathcal{S}}\{w_{0i} + qv_{0i} + (1+q)\bar{T}\}$, also require:

$$\omega_B d_p > q\gamma, \qquad h_O^{\max} > \frac{M_{\mathcal{S}}}{d_p P_-}\max\{1, \gamma/(qK)\}. \tag{41}$$

The latter bounds make the old estate floor and both owner caps slack in actual choices and in the comparison without the cash restriction. In that comparison, optimal cash expenditure is:

$$\frac{w+qv}{D}\left[1+\alpha\frac{L}{p}+\vartheta\frac{\chi+L\kappa}{\chi+p\kappa}\right]. \tag{42}$$

The bracket is at least $C_{\min}$. Condition (40) makes this expenditure strictly greater than w for every admissible rebate. Thus the original mortgage limit is strictly restrictive. The income condition says that households cannot bring enough of their future resources forward to finance their preferred current bundle. Logistic tastes give these endowments positive owner mass; their stationary predecessors supply the matched current old. Together with $\beta \geq q$, these primitive inequalities imply Proposition 1.

B Direct allocation: settlement and aggregate housing

At any date, increase a selected young owner's housing by ε and reduce the old donor's housing by the same amount. Keep consumption and estates fixed. Set:

$$\Delta a' = -P_{t+1}\varepsilon, \qquad \Delta a^e = qP_{t+1}\varepsilon, \tag{43}$$

and transfer $p_t\varepsilon$ from old to young. The young budget's extra cost is $q\Delta a' + (1+q\tau^p)P_t\varepsilon = p_t\varepsilon$. The old budget releases the same amount. The old estate is unchanged since $q^{-1}\Delta a^e - P_{t+1}\varepsilon = 0$; the young household's future resources are unchanged since $\Delta a' + P_{t+1}\varepsilon = 0$. Old saving finances young borrowing, so net external assets and existing creditor payments are unchanged. The young mortgage restriction may be violated, as permitted for the direct planner.

At stationarity, the same measure $N\pi^O(i)\,\mathrm{d}F(i)$ describes both owner cohorts. A positive eligible set has a positive subset with uniform strict margins on the mortgage, caps, and log arguments. The finite log welfare change therefore permits a common sufficiently small housing transfer on this subset. No equal-mass assumption on unrelated income groups is needed.

For the conditional housing optimum, let $\lambda_H > 0$ be the stock constraint's multiplier. The exact allocation with caps is:

$$s_i^* = \min\{h_{m_i}^{\max} - \kappa n_i, \alpha/\lambda_H\}, \qquad h_j^{2*} = \min\{\bar{h}_j^2, \gamma/\lambda_H\}, \tag{44}$$

where $\bar{h}_j^2 = h_{m_j}^{\max}$ for renters and $\bar{h}_j^2 = \min\{h_O^{\max}, e_j/P_{t+1}\}$ for owners with retained estates. The multiplier makes total housing equal $\bar{H}$. If no cap binds, this yields (18).

For the stationary aggregate comparison, let Q be the common probability measure of endowments and realized tenure in either cohort. When every old cap and estate floor is slack, the first-order conditions imply:

$$\alpha/s_i = \frac{\beta}{q}pm_i + \mu_i L_{m_i} + \eta_i^y \geq pm_i = \gamma/h_i^o, \tag{45}$$

where $L_O = L$, $L_R = p$, and $\eta_i^y \geq 0$ is the young housing-cap multiplier. If the uncapped planner allocation is feasible, then:

$$H_Y^* - H_Y^{\mathrm{eq}} = \frac{N}{\alpha+\gamma}\mathbb{E}_Q[\alpha h_i^o - \gamma s_i] > 0. \tag{46}$$

Strictness follows from the positive constrained-owner mass. At $\beta = q$ and with slack young caps, the integrand equals $\mu_i L_{m_i} s_i h_i^o$. This is an aggregate statement for the stated regime. With active planner caps, the preceding minimum formulas apply and this aggregate ordering has not been established generally.

C Transfer implementation and funding

Keep fertility and tenure committed. At the baseline let $\Lambda = 1/x$, $m = K/z$, and $\lambda = \beta m/q$. The first-order conditions are $\Lambda = \lambda + \mu$ and $\alpha/s = \lambda p + \mu L$, with $\mu > 0$. For (24)–(25), choose $\Delta a' = -\phi P\varepsilon/q$. The original budget and mortgage give:

$$\Delta c + q\Delta a' + (1 + q\tau^p)P\varepsilon = G, \quad q\Delta a' + \phi P\varepsilon = 0, \quad \Delta a' + P\varepsilon + J = 0. \tag{47}$$

The updated choices satisfy every first-order condition. For sufficiently small $\varepsilon > 0$, all strict margins persist. Strict concavity of the original log problem with affine constraints establishes the conditional optimum, including its original estate restriction.

Along this direction, write $A = \mathrm{d}x_\varepsilon/\mathrm{d}\varepsilon$ and $\rho_\varepsilon = \alpha x_\varepsilon/s_\varepsilon$. Then:

$$A = \frac{\rho_\varepsilon^2}{\alpha L} \geq \frac{L}{\alpha} \geq \frac{(1+\omega_B)p}{\gamma} = 1/g - p. \tag{48}$$

The first inequality uses $L \leq \rho_\varepsilon \leq p$, since $\phi \geq q$. The second is (22). Thus $R \geq 0$. Let m_F be the funders' mean marginal utility of cash. For one unit of recipient mass, the welfare derivative is:

$$\begin{aligned} \mathcal{W}'(0) &= \Lambda A + \alpha/s - m/g - m_F(A + p - 1/g) \\ &= (\Lambda - m)A + (\alpha/s - pm) + (m - m_F)(A + p - 1/g) > 0. \end{aligned} \tag{49}$$

The first two terms are strictly positive when $\beta \geq q$ and $\mu > 0$; the third is nonnegative. Integrate over recipients and collect their total residual equally across a positive funder group. Uniform strict margins and continuity give a finite welfare gain, even when the two groups have different masses.

All original mortgages are repaid. At date one the extra title held by the selected young is sold when they choose their unchanged old housing; these sales offset the smaller death sales of the initial old. The government's saving pays J, and its balance is zero. From date two the inherited household state is the baseline state. For each matched transfer, aggregate consumption and the initial old's estate payments satisfy:

$$\Delta C_0 = \frac{\omega_B \Delta x}{1+\omega_B}, \qquad \Delta E_1 = -\frac{\omega_B \Delta x}{q(1+\omega_B)}, \qquad \Delta C_0 + q\Delta E_1 = 0. \tag{50}$$

Thus the extra consumption is financed by smaller estates, with the old's loss included in welfare.

A primitive funder bound. Under Appendix A, put $\delta = \phi/q - 1 \geq 0$. A positive endowment group satisfying:

$$y_f^o > \delta P_+ h_O^{\max} + \frac{K d_p P_+ h_O^{\max}}{\gamma} \tag{51}$$

has strictly capped old owner housing. Since its cap threshold is $z_* = Kph_O^{\max}/\gamma$, its marginal utility is $m_f = (1+\omega_B)/(z_f - ph_O^{\max}) < \gamma/(ph_O^{\max})$. An uncapped donor has $z_i < z_*$ and hence $m_i = K/z_i$ above that threshold, which proves (23). This follows from $z_f \geq y_f^o - \delta P_+ h_O^{\max}$ and the old log demands. The price and rebate bounds use the full distribution, including the funders.

These restrictions are jointly possible. Choose $\phi > q$ close enough to q that $\ell > 1/(1+q)$, and choose $0 < \alpha < \gamma\ell/(1+\omega_B)$ with $\omega_B d_p > q\gamma$. For large ϑ, $k \to \ell^{-1} - 1 < q$. Choose $h_R^{\max} > \kappa/\nu$, large enough ϑ, and then small positive χ to satisfy (35). Let ordinary types have $w_0 = W$ and $v_0 = V_S$ with $qV_S > kW$. For fixed small $c_* > 0$, give a funder group old income $V > V_S + c_*$ and mass $c_*/(V - V_S)$, replacing that mass of ordinary types. Then $\bar{M}_0 = W + qV_S + qc_*$ is independent of V. Choose the finite owner cap from (41). Every bound on the right of (51) is now independent of V, so sufficiently large finite V satisfies it with positive funder mass. The restrictions are analytical sufficient conditions, not a claim of empirical mildness.

A simpler special case. If $\phi = q$ and $\alpha(1+\omega_B) = \gamma$, then $L = p$ and $J = R = 0$. A current grant $b > 0$ and an equal tax on the matched old suffice, without additional funders. Put $w_n = w - (\chi + p\kappa)n$; young adult choices are $x = w_n/(1+\alpha)$ and $s = \alpha w_n/[p(1+\alpha)]$. Both ages have housing response $g = \alpha/[p(1+\alpha)] = \gamma/(Kp)$ to cash. Welfare changes by:

$$\Delta\mathcal{W}(b) = (1+\alpha)\log(1 + b/w_n) + K\log(1 - b/v) > 0 \tag{52}$$

for sufficiently small $b > 0$, since strict finance and $\beta \geq q$ give $(1+\alpha)/w_n > K/v$. This is a useful exact illustration, but it requires two parameter equalities.

D Private fertility and the limits of the policy comparison

At fixed prices, let $\mathcal{V}(z)$ retain every old-age constraint. The reduced young problem is (14), with $L = p$ for renters. This old log problem is strictly concave, continuous, and piecewise smooth. When finance binds and young housing is uncapped, put $\Delta = L - p$ and $\Gamma = -\beta\mathcal{V}''(z)/q^2 > 0$. The curvature terms for choices (h, n) are:

$$\begin{aligned} A &= L^2/x^2 + \alpha/s^2 + \Gamma\Delta^2, \quad B_c = -L\chi/x^2 + \alpha\kappa/s^2, \\ D_n &= \chi^2/x^2 + \alpha\kappa^2/s^2 + \vartheta/n^2, \qquad J_n = AD_n - B_c^2 > 0. \end{aligned} \tag{53}$$

Implicit differentiation with respect to current cash yields:

$$h_w = \frac{LD_n + \chi B_c}{x^2 J_n} > 0, \qquad n_w = \frac{LB_c + \chi A}{x^2 J_n} > 0, \tag{54}$$

because $LD_n + \chi B_c = L\vartheta/n^2 + \alpha\kappa(\chi + L\kappa)/s^2 > 0$ and $LB_c + \chi A = \alpha(\chi + L\kappa)/s^2 + \chi\Gamma\Delta^2 > 0$. If young housing is capped and finance binds, $n_w = \chi/(x^2 D_n) > 0$.

If both young constraints are slack, expenditures on adult goods, adult space, and children have fixed positive shares of total current expenditure; that total rises with wealth by strict concavity of old utility. If only the young housing cap binds, put $k_c = \chi/(x^2 D_n)$. Then:

$$c_w = \frac{\Gamma}{\Gamma + (1 - \chi k_c)/x^2} > 0, \qquad n_w = k_c c_w > 0. \tag{55}$$

Here $0 < \chi k_c < 1$. At changes in old constraint regimes, $\mathcal{V}'$ is continuous and the one-sided responses remain positive. Unique choices join continuously at all regime changes, extending monotonicity to finite gifts. Housing is strictly increasing whenever its young cap is slack.

For the finite test, define $f(n; c, h) = \vartheta/n - \chi/(c - \chi n) - \alpha\kappa/(h - \kappa n)$. This function strictly decreases in n over its feasible interval, and the chosen fertility is its unique zero. Evaluating the new function at n_0 gives (28). If $c_1 \leq \chi n_0$ or $h_1 \leq \kappa n_0$, the new bundle cannot accommodate n_0, so $n_1 < n_0$.

For a current transfer $\mathrm{d}b$ with future repayment of present value $r\,\mathrm{d}b$, with finance binding and young housing uncapped, the calculation gives:

$$\frac{\mathrm{d}n}{\mathrm{d}b} = \frac{(LB_c + \chi A)/x^2 + r\Gamma\Delta B_c}{J_n}. \tag{56}$$

The gift has $r = 0$; the term involving repayment can be negative. Similarly, for a common gift before tenure choice, its derivative includes the term $\pi^O(1 - \pi^O)(n^O - n^R)(1/x_O - 1/x_R)/\sigma_\xi$, whose sign is not fixed. These are reasons to specify the policy before inferring aggregate fertility from the individual gift result.

One could instead redesign the two grants to hold old resources fixed while fertility adjusts. Along the direction in (24), both adult goods and adult space rise, and:

$$\mathrm{d}n = \frac{n^2}{\vartheta}\left(\frac{\chi}{x^2}\mathrm{d}x + \frac{\alpha\kappa}{s^2}\mathrm{d}s\right) > 0. \tag{57}$$

The redesigned payments and residual funding would be:

$$\begin{aligned} G &= \Delta x + L\varepsilon + (\chi + L\kappa)\Delta n, \\ qJ &= (p - L)(\varepsilon + \kappa\Delta n), \\ R &= \Delta x + (p - 1/g)\varepsilon + [\chi + \kappa(p - 1/g)]\Delta n. \end{aligned} \tag{58}$$

This is a different policy from holding the original grant pair fixed. Its corrected funding term matters even at unchanged prices. In the simple case $L = p$ and $\alpha(1 + \omega_B) = \gamma$, its derivative along an adult-space increase is $R' = (\chi - \kappa p/\alpha)n'$. It is negative if $\chi/\kappa < p/\alpha$, although the fixed-fertility funding condition holds at equality. Reusing that proof with free fertility would therefore miss a current funding term as well as the change in future population.

E Voluntary fertility with unchanged cohort size

This construction removes the commitment restriction in Proposition 3, at the cost of a more specific policy and $\phi = q$. The intervention is unexpected, announced after the ownership taste draw and before the young make any choices. The authority can select households using predetermined identities, endowments, and their realized ownership tastes. Selected young households strictly prefer ownership at baseline. Transfers apply regardless of their subsequent tenure choice, so both tenure and fertility remain freely chosen. This targeting requires more information than an anonymous age-transfer rule.

Put $H_O = h_O^{\max}$ and $E = 1 + \alpha + \vartheta$. Since $\phi = q$, we have $L = p$, and a strictly constrained young owner's old resources equal v. For an uncapped young owner, a unit of cash raises fertility by a_n and housing by a_h:

$$a_n = \frac{\vartheta}{E(\chi + p\kappa)}, \qquad a_h = \frac{\alpha/p + \vartheta\kappa/(\chi + p\kappa)}{E}, \qquad g = \frac{\gamma}{Kp}. \tag{59}$$

Select equal positive submeasures of four groups: an uncapped, constrained young owner A; a constrained young owner B strictly at H_O; an uncapped old counterpart of A, with resources v_A and slack estate floor; and an old owner strictly at H_O with slack estate floor. The latter can be B's stationary counterpart. Uniform strict ownership margins for the young preserve their voluntary tenure choices under small transfers.

At B's cap, let $x_B = w_B - pH_O - \chi n_B$ and $s_B = H_O - \kappa n_B$. Its cash response of fertility is $b_B > 0$. Define the tax per unit grant needed to offset births, q_B, and the old tax needed to release housing, d:

$$b_B = \frac{\chi/x_B^2}{\vartheta/n_B^2 + \chi^2/x_B^2 + \alpha\kappa^2/s_B^2}, \qquad q_B = \frac{a_n}{b_B}, \qquad d = \frac{a_h}{g}. \tag{60}$$

A sufficient condition for the construction is:

$$0 < q_B < 1, \qquad q_B + d > 1, \qquad \frac{E}{w_A} > \frac{q_B}{x_B} + \frac{Kd}{v_A}. \tag{61}$$

The last inequality compares the young recipient's gain per unit of cash with the losses of the two taxpayers. The primitive construction below makes these restrictions explicit and shows that they can hold jointly. There is no additional restriction on β/q.

Give young A a small grant G. Tax young B by the amount $Q(G)$ satisfying:

$$n_B(w_B - Q(G)) = n_B(w_B) - a_n G. \tag{62}$$

This uses B's privately optimal fertility after the tax. Since $b_B > 0$, the small tax is unique and $Q'(0) = q_B$. Tax old A by dG and give the capped old group $S(G) = Q(G) + dG - G > 0$. The program balances immediately: $G + S = Q + dG$. A's housing rises by $a_h G$, B stays at its cap, old A's housing falls by $gdG = a_h G$, and the capped old recipients retain their housing. Thus the original price clears housing. Births also remain unchanged.

For both selected young owners, the mortgage satisfies $a' = -Ph$, so $a' + Ph + v = v$ is unchanged. They therefore make their original old-age choices. Current young of other types and all future entrants face the same prices and resources, while their cohort masses are unchanged. A's larger inherited title offsets its larger mortgage, including at resale. Initial old consumption and estate payments change, with $\Delta C_0 + q\Delta E_1 = 0$ by the original budgets and the balanced transfer account. No new government debt or unfinanced external resources are needed.

Let $m_F > 0$ be the capped old recipient's marginal utility of cash. The welfare derivative, including both young households' fertility choices, is:

$$\mathcal{W}'(0) = \frac{E}{w_A} - \frac{q_B}{x_B} - \frac{Kd}{v_A} + m_F(q_B + d - 1) > 0. \tag{63}$$

The first three terms are positive in sum by (61), and the last is positive. Strict regime and ownership margins and continuity give a finite positive transfer. Thus this is an equilibrium utilitarian improvement that moves housing toward the young with the original choice timing. Aggregate fertility is held fixed through voluntary offsetting responses, rather than by fixing individual fertility. Tenure probabilities must be computed from individual choice inequalities and the original taste distribution. The endowment-only logit formula does not apply to taste-dependent transfers. Here all tenure choices remain at baseline.

An explicit cap interval. Put $r = \chi/(p\kappa)$, $a = \alpha + \vartheta$, and define:

$$f(t) = t + \frac{t(1-t)}{\vartheta - at}, \quad t_* = \frac{\vartheta}{\alpha(r+1)+\vartheta}, \quad t_\dagger = \frac{\vartheta}{a}\left[1 - \sqrt{\frac{\alpha r}{E(\alpha r + a)}}\right]. \tag{64}$$

If $\alpha r > 1$, then $t_* < t_\dagger < \vartheta/a$. Choose $t_B \in (t_*, t_\dagger)$ and set $w_B = pH_O + (\chi H_O/\kappa)f(t_B)$ and $n_B = H_O t_B/\kappa$. These solve B's fertility condition with a strict cap. Here $f' > 0$, $f'' > 0$, and $q_B = a_n\chi f'(t_B) < 1$. At the lower endpoint:

$$q_* = 1 - pa_h\left(1 - \frac{1}{\alpha r}\right), \qquad q_* + d - 1 = a_h\left(\frac{1}{g} - p + \frac{p}{\alpha r}\right) > 0. \tag{65}$$

Consequently the first two conditions in (61) hold throughout this explicit interval. The finite tax is:

$$Q(G) = \frac{\chi H_O}{\kappa}\left[f(t_B) - f\left(t_B - \frac{\kappa a_n G}{H_O}\right)\right]. \tag{66}$$

For $G < H_O(t_B - t_*)/(\kappa a_n)$, B remains strictly capped and $0 < Q'(G) < 1$, $S'(G) > 0$. Further restrict G by A's housing cap and mortgage margin, old A's positive resources, and the ownership margins.

A nonempty family of primitive economies. Fix any finite $\beta > 0$, $q \in (0, 1)$, and positive p, H_O and preference parameters with $\alpha\chi > p\kappa$ and $\omega_B(1 - q) > q\gamma$. Choose $\eta \in (0, 1)$ and set $v_A = \eta H_O/g$; choose B's cash from the interval just given. The following inequalities are sufficient:

$$\begin{gathered} v_B > \max\{H_O/g,\ pH_O + \beta(1 + \omega_B)x_B/q\}, \\ 0 < w_A < \min\left\{H_O/a_h,\ qEv_A/(\beta K),\ E/(q_B/x_B + Kd/v_A)\right\}. \end{gathered} \tag{67}$$

They make old B capped, both young mortgages strict, young A uncapped, and the welfare inequality strict. Every upper bound on w_A is positive. Give the two endowment types positive mass, choose any $0 < h_R^{\max} < H_O$, and retain logistic tastes with positive scale. Both tenure menus have finite values, so both tenures occur and positive subsets of strict owners can be selected.

To close the stationary economy, evaluate these menus at the auxiliary total resources (w_i, v_i) and service price p. Let $\bar{h}_{\text{life}}$ be mean lifetime housing. Choose a small positive property tax, set $P = p/(1 - q + q\tau^p)$, and set:

$$T = q\tau^p P\bar{h}_{\text{life}}/2, \qquad N = \bar{H}/\bar{h}_{\text{life}}. \tag{68}$$

The bound $\tau^p < (1 - q)\min_i\{w_i, v_i\}/(2qpH_O)$ ensures that T is smaller than every chosen resource level. Set primitive current income plus wealth to $w_i - T$, split into positive components, and old income to $v_i - T$. All real menus remain unchanged: p, L, w_i, v_i are unchanged, and the old estate floor remains slack.

For any given ν, multiply both provisional child costs by $\nu\bar{n}_0$, where $\bar{n}_0$ is mean fertility before this rescaling. Fertility is divided by the same factor, leaving goods and space used by children, all other real choices, and tenure-value differences unchanged. Mean fertility becomes $1/\nu$. The young choices generate the stationary old distribution, completing equilibrium. This is an analytical construction of primitive economies satisfying the theorem, not a claim about already fixed calibration parameters or a general mortgage share.

F Welfare weights and the retained Pareto results

The consumption-log coefficients are one at both ages. Under the benchmark in (15), each living household receives equal weight on remaining utility. If instead the initial old receive weight $\rho > 0$, the stationary direct gap is:

$$\alpha/s - \rho\gamma/h^2 = (\beta/q - \rho)pm + \mu L. \tag{69}$$

Equal weights on lifetime utility measured at birth imply $\rho = \beta$, since old utility carried that factor in the original lifetime objective. The condition then differs from the remaining-utility convention $\rho = 1$. Choosing between them is a welfare judgment, not a harmless change in notation.

The earlier note, *Housing and Fertility with Conventional Mortgage Finance*, retains the separate compensated Pareto constructions and their stronger assumptions. Those results can serve as appendix material. The present utilitarian argument does not use them and does not relabel its uncompensated gains as Pareto improvements. Comparing policies that change who is born would additionally require a population-welfare criterion; the fertility and population statements here make no such ranking.